# Search-Complexity Analysis of an Embedded Zero-Vector Penalty

A paired study of leaf-cost equivalence, partial metrics, pruning margins, and sphere-decoder search paths

**Thesis Analysis Note**

This document isolates the comparison between an explicit zero-vector branch penalty and its quadratic embedding in the sphere-decoder matrix. It records the mathematical equivalence, the observed complexity difference, the resulting hypotheses, and the reproducible diagnostic methodology. Its purpose is to provide a rigorous source for later integration into the master thesis; it is not a substitute for the final thesis chapter.

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# 1 Problem Statement and Experimental Observation

The long-horizon finite-control-set model predictive controller uses a binary sphere decoder to determine a switching sequence

$$\boldsymbol{U} = [S_{a,1}, S_{b,1}, S_{c,1}, \ldots, S_{a,N_p}, S_{b,N_p}, S_{c,N_p}]^{\mathsf{T}}, \qquad \boldsymbol{U} \in \{0,1\}^{3N_p}, \tag{1.1}$$

with $N_p = 5$ and hence 15 binary decision variables. Two mathematically equivalent treatments of the zero vectors 000 and 111 were implemented. The first adds a discrete penalty when a complete three-switch block is known. The second incorporates an equivalent quadratic expression into the Hessian before its Cholesky factorization.

Paired simulations at 10, 50, and 90 Nm produced the following initially surprising observations.

1. The two formulations returned identical optimal switching sequences in all paired samples considered.
2. The embedded formulation reported an optimal cost approximately 6250 larger than the explicit formulation at every transaction.
3. The embedded formulation visited substantially more search nodes.
4. Babai initialization was the best of the tested initializers in both formulations and dominated especially strongly in the embedded formulation.

These results separate control equivalence from computational equivalence. An identical minimizer at the leaves of a search tree does not imply identical partial metrics or identical pruning decisions inside the tree.

# 2 Leaf-Level Equivalence of the Two Penalties

## 2.1 Exact binary representation

For one three-phase switching vector define

$$m = S_a + S_b + S_c, \qquad m \in \{0, 1, 2, 3\}. \tag{2.1}$$

The vector is a zero vector exactly when $m = 0$ or $m = 3$. The explicit cost is

$$J_{\mathrm{z,exp}}(m) = \begin{cases} \lambda_z, & m \in \{0, 3\}, \\ 0, & m \in \{1, 2\}. \end{cases} \tag{2.2}$$

On the binary domain this logic is represented exactly by

$$J_{\mathrm{z,exp}}(m) = \frac{\lambda_z}{2}(m-1)(m-2). \tag{2.3}$$

The centered quadratic used by the embedded formulation is

$$J_{\mathrm{z,emb}}(m) = \frac{\lambda_z}{2}(m-1.5)^2. \tag{2.4}$$

Consequently,

$$\boxed{J_{\mathrm{z,emb}}(m) = J_{\mathrm{z,exp}}(m) + \frac{\lambda_z}{8}.} \tag{2.5}$$

For $\lambda_z = 10{,}000$, the explicit costs of zero and active vectors are 10,000 and zero, whereas the embedded costs are 11,250 and 1,250. Both therefore retain the same zero-to-active cost difference of 10,000. The embedded representation merely adds 1250 to every complete switching vector. Over $N_p = 5$ steps,

$$C = \frac{N_p \lambda_z}{8} = 6250. \tag{2.6}$$

For every complete binary sequence,

$$J_{\mathrm{emb}}(\boldsymbol{U}) = J_{\mathrm{exp}}(\boldsymbol{U}) + C, \tag{2.7}$$

and therefore both formulations have the same minimizer.

## 2.2 Why leaf equivalence does not prove search equivalence

The SDA does not wait until all 15 variables are known. With the lower triangular factor used by the implemented forward search, its quadratic partial metric at depth $i$ is

$$P_i^{\mathrm{quad}} = \sum_{k=1}^{i} \left( (HU_{\mathrm{unc}})_k - \sum_{j=1}^{k} H_{kj} U_j \right)^2 . \tag{2.8}$$

A branch is pruned when its complete partial metric exceeds the current incumbent radius $R$:

$$P_i > R. \tag{2.9}$$

Equation (2.6) constrains the leaf cost at $i = 15$, but it does not imply that partial metrics at $i < 15$ differ by a fixed or uniformly distributed fraction of 6250.

# 3 Search Mechanism and Working Hypotheses

## 3.1 Explicit block-boundary penalty

The production search proceeds from $i = 1$ to $i = 15$. At $i = 3, 6, 9, 12, 15$, a complete three-phase vector is available. The explicit formulation then evaluates the zero-vector condition and, if necessary, adds $\lambda_z$ immediately. This creates a semantic alignment between the physical event “a zero vector has been completed” and a discrete jump in the partial search metric.

## 3.2 Embedded quadratic representation

Using $S_x^2 = S_x$ for binary variables, the embedded block cost expands to

$$J_{\mathrm{z,emb}} = \frac{9\lambda_z}{8} - \lambda_z(S_a + S_b + S_c) + \lambda_z(S_aS_b + S_aS_c + S_bS_c). \tag{3.1}$$

It is therefore not composed of three independent one-third penalties. Its final classification is produced jointly by linear and pairwise terms. After the complete Hessian is factorized,

$$\boldsymbol{W} = \boldsymbol{H}\boldsymbol{H}^{\mathsf{T}}, \tag{3.2}$$

the original block meaning is generally redistributed among several squared residuals. The individual residual increment at one search level is thus not itself interpretable as a fraction of the zero-vector penalty.

## 3.3 Refined prefix-slack hypothesis

For the same binary prefix, define the effective accumulated difference

$$C_i^{\mathrm{eff}} = P_i^{\mathrm{emb}} - P_i^{\mathrm{exp}} \tag{3.3}$$

and the remaining representation-induced prefix slack

$$S_i = C - C_i^{\mathrm{eff}}. \tag{3.4}$$

If the two initial radii differ by $C$, the pruning margins satisfy

$$M_i^{\mathrm{exp}} = R_{\mathrm{exp}} - P_i^{\mathrm{exp}}, \tag{3.5}$$

$$M_i^{\mathrm{emb}} = R_{\mathrm{emb}} - P_i^{\mathrm{emb}}, \tag{3.6}$$

$$M_i^{\mathrm{emb}} - M_i^{\mathrm{exp}} = S_i. \tag{3.7}$$

A positive $S_i$ means that the embedded representation leaves more room before the same prefix reaches its pruning boundary. At a complete leaf, $C_{15}^{\mathrm{eff}} = C$ and $S_{15} = 0$ are expected. At intermediate depths, neither quantity is assumed to be uniform or path independent.

The constant 6250 is not random noise. It is a deterministic common offset. Nor is a larger numerical radius alone sufficient to explain additional nodes: adding the same constant consistently to every bound and every radius would leave all pruning decisions unchanged. The proposed mechanism is the possible mismatch between the offset already present in a complete incumbent and the amount represented in a shallow partial metric.

## 3.4 Evidence status

Three statements must remain distinct.

1. **Proved:** the complete binary objectives differ by the constant $C = 6250$.
2. **Observed:** optimal switching sequences coincide while the embedded search visits more nodes.
3. **Hypothesized mechanism:** factorization changes the tightness and depth distribution of prefix bounds, delaying pruning for many prefixes.

The third statement requires same-prefix analysis and must not be presented as a proven causal conclusion solely from row norms of $H$.

# 4 Diagnostic Quantities and Their Interpretation

## 4.1 Paired node difference

For transaction $k$, paired by operating condition and controller time, define

$$\Delta N_k = N_{\mathrm{emb},k} - N_{\mathrm{exp},k}. \tag{4.1}$$

Thus $\Delta N_k > 0$ means that the embedded solver visited more nodes for the same physical transaction. Pairing is essential: two unpaired node-count samples may correspond to different motor states and are not a controlled comparison.

The plotted probability-density curves are kernel-density estimates (KDEs):

$$\widehat{f}_h(x) = \frac{1}{nh} \sum_{k=1}^{n} K\left( \frac{x - \Delta N_k}{h} \right), \tag{4.2}$$

where $K$ is a normalized smoothing kernel and $h$ is its bandwidth. The vertical scale is probability density, not sample probability: the area under each curve equals one. A curve concentrated to the right of zero indicates that positive node differences dominate the paired population.

## 4.2 Visited, pruned, and surviving nodes by depth

Let $N_i^{\mathrm{vis}}$ be the number of candidates evaluated at depth $i$, $N_i^{\mathrm{pr}}$ the number rejected by (2.9), and $N_i^{\mathrm{sur}}$ the number allowed to descend. The depth-wise pruning rate is

$$p_i^{\mathrm{prune}} = \frac{N_i^{\mathrm{pr}}}{N_i^{\mathrm{vis}}}. \tag{4.3}$$

Peaks at $i = 3, 6, 9, 12, 15$ are of particular interest because these are the three-switch block boundaries.

## 4.3 Pruning margin $R - P_i$

For every visited node, the pruning margin is defined as

$$M_i = R - P_i. \tag{4.4}$$

Here $R$ is the incumbent radius immediately before the node is tested, not necessarily the initial radius. Its interpretation is:

$$M_i > 0 : \text{node remains inside the sphere}, \qquad M_i < 0 : \text{node is pruned}. \tag{4.5}$$

The median and interquartile range summarize the path-dependent distribution at each depth. A large positive margin does not mean that a candidate is good; it only means that the current lower bound has not yet excluded it.

## 4.4 Dynamic incumbent radius

The incumbent radius can shrink whenever a better complete leaf is found. Plotting $R$ against visited-node index distinguishes two effects:

1. initialization quality, which determines the first radius; and
2. discovery order, which determines how quickly subsequent leaves tighten the radius.

Consequently, comparing initial radii alone is insufficient.

## 4.5 Node-level trace

The independent profiler records the search depth, binary candidate, encoded prefix, incoming metric, squared residual, zero-vector increment, dead-time increment, complete partial metric, radius, pruning margin, pruning decision, leaf flag, and radius update. Zero-vector and dead-time increments are stored separately so that a block-boundary rejection can be assigned to its actual cause.

# 5 Reproducible Experimental Design

The SDI archive contains 40,000 SDA transactions for each run. The analysis extracts all inputs and reference outputs and replays them with an independent MATLAB implementation of the production search. The replay is accepted only when $J_{\mathrm{opt}}$, the node count, and $\boldsymbol{U}_{\mathrm{opt}}$ match the logged reference.

Two complementary analyses are required.

1. **Actual-search replay** records the nodes genuinely visited by Schnorr–Euchner enumeration, including dynamic radius updates.
2. **Complete same-prefix enumeration** evaluates both formulations for the same $(i, \text{prefix})$ pairs. It avoids the selection bias that would result from comparing the $k$th visited node of two different search paths.

Representative transactions are selected reproducibly from the distribution of $\Delta N$, rather than chosen visually. For each torque condition, the cases nearest the 5th, 50th, and 95th percentiles are used. The 50th percentile represents typical behavior, the 95th percentile represents a pronounced complexity increase, and the 5th percentile acts as a weak-difference control.

# 6 Current Results

## 6.1 Population-level comparison

The full paired data show that the node-difference distributions are located predominantly to the right of zero, while the leaf-cost difference agrees with 6250 to numerical precision.

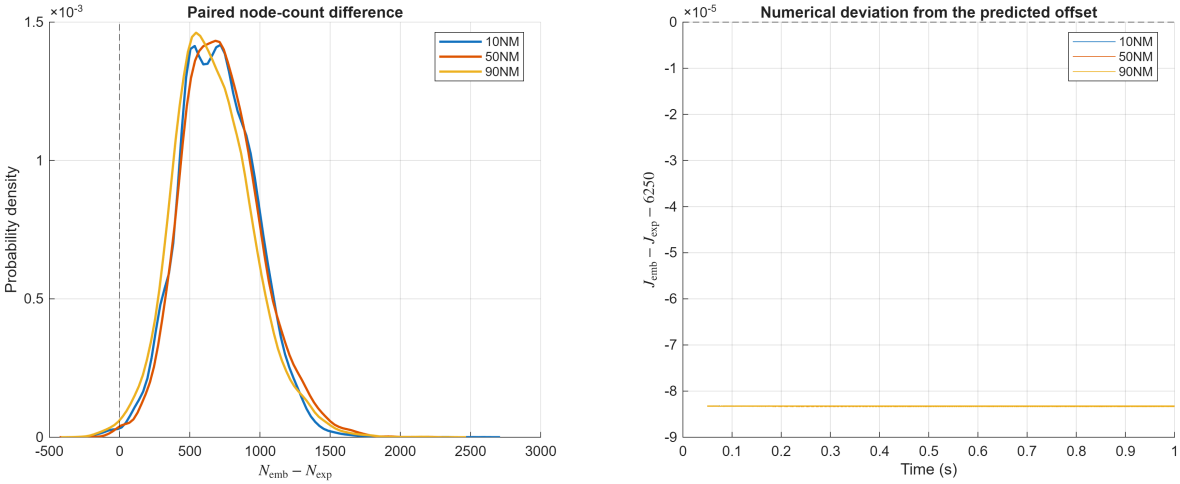



Figure 6.1: Left: kernel-density estimate of the paired node-count difference $\Delta N$. The area below each curve equals one. Right: numerical deviation of the measured leaf-cost difference from the predicted constant 6250.

Independent replay of 500 uniformly distributed transactions per run produced the mean node counts in table 6.1. Every replayed transaction matched the logged optimum, node count, and switching sequence.

Table 6.1: Mean visited nodes in the independently replayed sample.

| Torque | Embedded $H$ | Explicit penalty | Difference |
|---|---|---|---|
| 10 Nm | 1151.87 | 479.55 | 672.32 |
| 50 Nm | 1177.64 | 474.69 | 702.95 |
| 90 Nm | 1095.46 | 462.34 | 633.12 |

## 6.2 Depth-resolved behavior

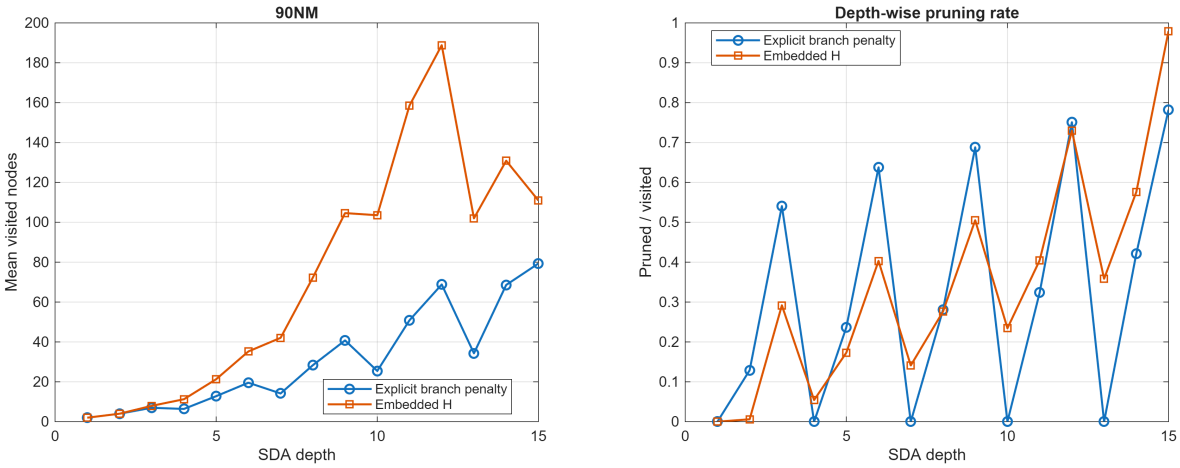



Figure 6.2: Mean visited nodes and aggregate pruning rate versus search depth at 90 Nm. The explicit formulation exhibits pronounced pruning-rate peaks at several complete three-switch block boundaries, while the embedded formulation carries more surviving candidates into deeper levels.

The same qualitative behavior appears at 10 and 50 Nm. This result supports the block-boundary-alignment hypothesis, but does not yet prove that the 6250 offset is the sole cause.

## 6.3 Representative median case

At 90 Nm, the median node-difference case occurs at transaction 2305 and $t = 0.057625$ s. The explicit search visits 494 nodes and the embedded search 1140 nodes, giving $\Delta N = 646$.

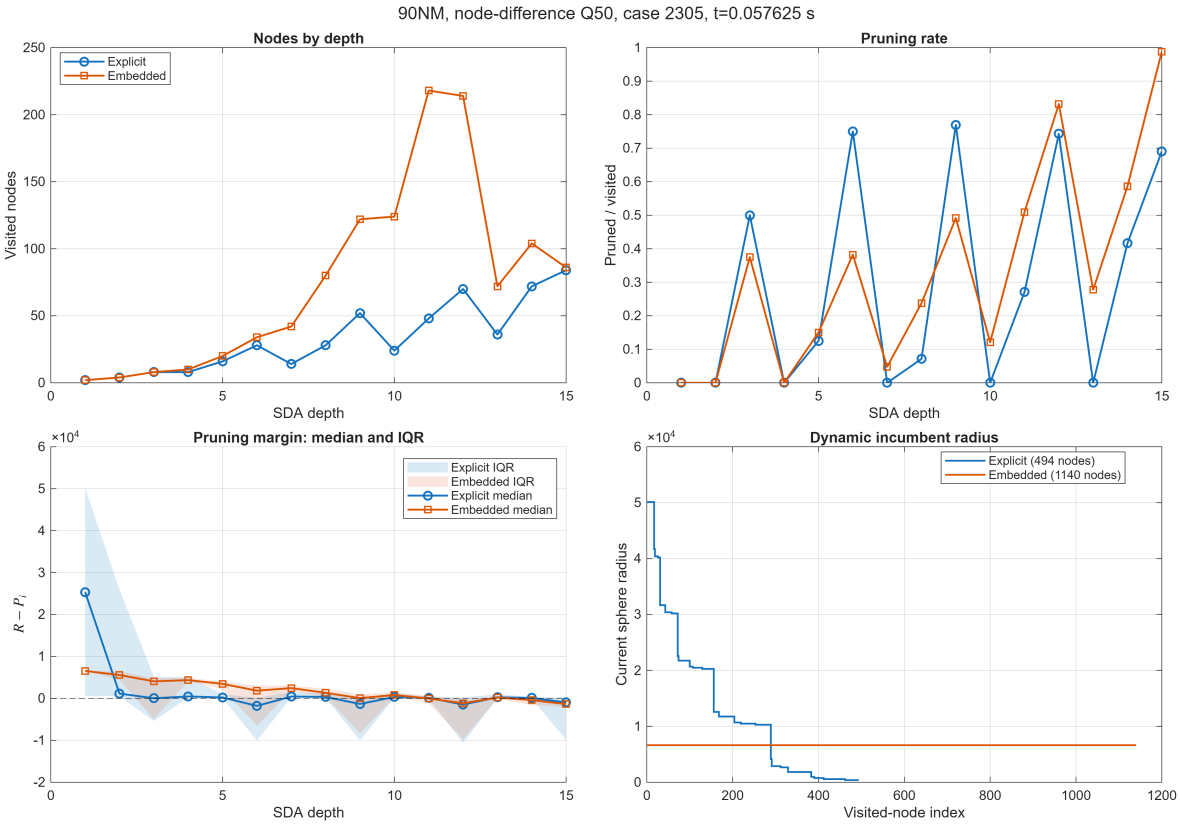



Figure 6.3: Search diagnostics for the 90 Nm median node-difference case. The lower-left panel reports the dynamic pruning margin from equation (4.4); shaded regions are interquartile ranges. The lower-right panel shows the incumbent radius immediately before each visited node.

The figure adds an important qualification to the initial hypothesis. In this case the embedded Babai incumbent is already close to its final value and its radius remains almost constant. The explicit formulation starts from a much larger incumbent but discovers better leaves early and rapidly contracts its radius. The embedded solver nevertheless visits more nodes, especially at intermediate depths. Hence the raw initial-radius magnitude cannot by itself explain the complexity difference. The evidence instead points to the joint effect of prefix-bound tightness, block-boundary penalties, enumeration order, and dynamic incumbent updates.

## 6.4 Representative-case definition

The selected $(Q_5, Q_{50}, Q_{95})$ node differences are respectively $(282, 692, 1148)$ at 10 Nm, $(314, 706, 1226)$ at 50 Nm, and $(244, 646, 1166)$ at 90 Nm. All corresponding figures and

exact transaction indices are retained with the analysis data to prevent subjective figure selection.

## 6.5 Complete same-prefix enumeration

The complete prefix tree was subsequently enumerated for all nine selected transactions. At depth $i$, all $2^i$ binary prefixes were evaluated under both formulations. This removes the ambiguity of comparing differently ordered, actually visited nodes. At every complete leaf and in every selected case, the measured difference was

$$P_{15}^{\mathrm{emb}} - P_{15}^{\mathrm{exp}} = 6249.99992 \tag{6.1}$$

for all $2^{15}$ sequences; the tiny deviation from 6250 is numerical.

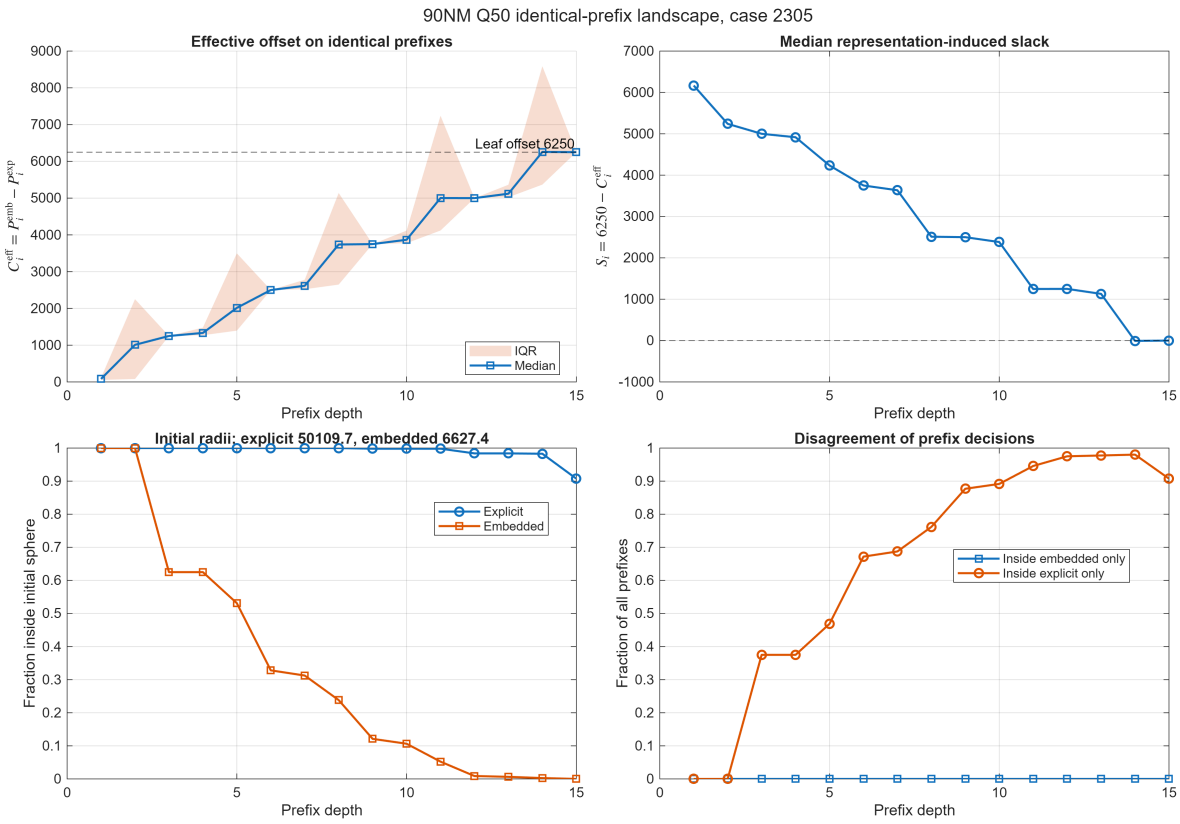



Figure 6.4: Complete same-prefix landscape for the 90 Nm median case. Top left: effective prefix offset and its interquartile range. Top right: remaining median slack. Bottom left: fraction of all prefixes inside each formulation's initial sphere. Bottom right: prefixes classified inside only one of the two initial spheres.

The median $C_i^{\mathrm{eff}}$ increases progressively toward 6250, and the corresponding median

$S_i$ decreases toward zero. This confirms that the leaf offset is not uniformly present in shallow partial metrics. However, the initial-radius comparison provides an important counterexample to a simple “larger embedded sphere” explanation. In this transaction,

$$R_{0,\mathrm{exp}} = 50109.7, \qquad R_{0,\mathrm{emb}} = 6627.4. \tag{6.2}$$

The explicit initial sphere contains almost all prefixes through most depths, whereas the embedded initial sphere is substantially tighter. A static test against the initial radii would therefore predict more aggressive embedded pruning, opposite to the actually observed node count.

The discrepancy is resolved by recognizing that the initial sphere is not the sphere used throughout the search. As shown in figure 6.3, the explicit search discovers progressively better leaves and rapidly reduces its incumbent radius. The embedded Babai incumbent begins near its final value and remains almost unchanged. Actual complexity is consequently determined by the coupled sequence

$$\text{conditional centers} \rightarrow \text{enumeration order} \rightarrow \text{time of first good leaves} \rightarrow \text{dynamic radius contraction} \rightarrow \tag{6.3}$$

in addition to the depth distribution of the zero-vector cost.

# 7 Remaining Analysis

The current results prove leaf equivalence, establish a repeatable complexity difference, and complete the same-prefix comparison of $P_i^{\mathrm{exp}}$, $P_i^{\mathrm{emb}}$, $C_i^{\mathrm{eff}}$, $S_i$, and initial-sphere membership. They also reject the hypothesis that a larger embedded initial radius is by itself responsible for the additional nodes.

The remaining analysis should reconstruct the event sequence of the dynamic search. For every radius update it should record the leaf that caused the update, the number of nodes visited before that update, and the number of subsequent prefixes rejected under the new radius. A useful counterfactual is to replay each formulation with a fixed common incumbent sequence and then with dynamic updates enabled. This separates representation-dependent prefix bounds from representation-dependent discovery order.

The refined causal chain to test is

$$\text{quadratic representation and conditional centers} \Longrightarrow \text{different enumeration order and leaf discovery t} \tag{7.1}$$

The block-boundary penalty and nonuniform prefix offset remain contributing mechanisms, but the evidence no longer supports attributing the observed node difference to the raw initial-radius magnitude alone.

# 8 Reproducibility Assets

The analysis source, extracted data, validation outputs, representative-case selection, and generated figures are maintained under `Model_2025b/SDA_embedded_analyze`. The production Simulink model and production SDA source are not modified by the profiling workflow.